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Abstract

Mass spectrometry is a cornerstone technique in analytical chemistry, metabolomics, pharmaceutical research and many other areas. The ability to predict a molecular structure from a given spectrum, and vice versa, remains a central challenge. While forward prediction of spectra from known structures is increasingly accurate due to machine learning methods, the inverse problem — predicting possible molecular structures from a given spectrum — remains difficult, ambiguous, and poorly interpretable.

The work proposes the integration of aiding intermediate fragments, which are generated by graph transformation tools, as a strategy to enhance machine learning-based bidirectional translation between molecules and spectra. The aim is to enhance model interpretability and accuracy.

The work will offer a systematic method for integrating graph-based intermediates into machine learning pipelines, potentially improving interpretability and performance of structure $\leftrightarrow$ spectrum translation models.

Zusammenfassung

Massenspektrometrie ist eine grundlegende Technik in der Analytischen Chemie, der Metabolomik, der pharmazeutischen Forschung und in vielen weiteren Bereichen. Die Vorhersage einer Molekülstruktur aus einem gegebenen Spektrum — und umgekehrt — bleibt eine zentrale Herausforderung. Während die Vorhersage von Spektren aus bekannten Strukturen dank Methoden des maschinellen Lernens zunehmend präzise gelingt, ist das inverse Problem — also die Vorhersage möglicher Molekülstrukturen aus einem gegebenen Spektrum — weiterhin schwierig, mehrdeutig und nur schwer interpretierbar.

Die Arbeit schlägt vor, unterstützende Zwischenfragmente, die durch graphbasierte Transformationstools erzeugt werden, als Strategie zu integrieren, um das maschinelle Lernen für die bidirektionale Übersetzung zwischen Molekülen und Spektren zu verbessern. Ziel ist es, die Interpretierbarkeit und Genauigkeit der Modelle zu steigern.

Die Arbeit wird eine systematische Methode zur Integration graphbasierter Zwischenstufen in Machine-Learning-Pipelines bieten, um potenziell die Interpretierbarkeit und Leistungsfähigkeit von Struktur $\leftrightarrow$ Spektrum-Übersetzungsmodellen zu verbessern.

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Chapter 1

Introduction

1.1 Motivation

Mass spectrometry (MS) is one of the most widely used analytical techniques for molecular characterization across chemistry, metabolomics, environmental sciences, and pharmaceutical research. [1, p. 1] Its high sensitivity [2, p. 1764], reproducibility [2, p. 1764], and scalability [2, p. 1764] have enabled large-scale molecular profiling in both research and industrial settings. Modern instrumentation such as Orbitrap, Q-TOF, and triple quadrupole mass spectrometers generates high-throughput spectral data that can be compared against large repositories of molecular fingerprints such as the NIST Mass Spectral Library to support compound identification and profiling [3, pp. 7274-7275].

Despite these advances in data acquisition, the interpretation of MS data remains a significant bottleneck [4, p. 1]. Structural elucidation from spectra is still largely expert-driven, relying on chemical intuition and established fragmentation rules [4, p. 11]. While thousands of spectra can be recorded within a short time frame [4, p. 1], the corresponding structural annotation process remains comparatively slow, manual, and difficult to scale [4, p. 3]. This imbalance creates a fundamental asymmetry between data generation and data analysis.

The intrinsic complexity of molecular fragmentation contributes substantially to this challenge [1, p. 240]. Fragmentation processes are governed by chemically valid bond cleavages and rearrangements [1, pp. 30-31], leading to multiple competing fragmentation pathways. A single molecule may therefore generate numerous fragments through direct cleavages and rearrangement reactions [1, p. 240-241]. Because fragmentation pathways depend sensitively on molecular structure and internal energy [1, p. 26, pp. 30-31], structurally distinct molecules may produce partially overlapping spectral signatures, while small structural variations can induce significant

spectral differences. Consequently, the inverse problem — inferring molecular structure from spectral data — is inherently ill-posed [1, p. 240].

Recent machine learning approaches aim to address this challenge by learning direct mappings between molecular representations and spectral outputs. Graph neural networks [5], sequence-based encoders [6, p. 1573], and generative models have demonstrated promising performance in forward (molecule-to-spectrum) and inverse (spectrum-to-molecule) prediction tasks. However, purely base-data-driven approaches exhibit several limitations:

- They often lack mechanistic interpretability. [7, p. 3& 6]
- They do not explicitly model fragmentation pathways. [7, p. 3]
- They rely on statistical correlations rather than chemically structured priors. [8, p. 966]
- Their generalization to rare or out-of-distribution molecular scaffolds remains limited. [9, p. 12]

In particular, models that treat molecules as flat graphs or string encodings and spectra as unordered peak lists ignore the structured chemical processes underlying fragmentation. This omission reduces interpretability and may hinder robust extrapolation beyond the training distribution. [10, p. 3420]

Fragmentation in mass spectrometry is not an arbitrary stochastic process; it follows chemically constrained transformation sequences. [11, ch. 4& 8] Therefore, incorporating structured representations of fragmentation—such as derivation graphs encoding valid transformation rules—offers a principled way to introduce chemical inductive bias into machine learning models. By explicitly modeling fragment lineage and transformation pathways, it becomes possible to align molecular substructures with spectral evidence in a mechanistically meaningful manner.

This thesis is motivated by the hypothesis that integrating chemically structured derivation graphs into a machine learning framework can improve predictive robustness, interpretability, and bidirectional structure-spectrum modeling. By combining fragment-aware representations with data-driven learning, the aim is to bridge the gap between high-throughput spectral acquisition and scalable, chemically grounded interpretation.

1.2 Objectives

The objective of this thesis is to design, implement, and systematically evaluate a fragment-augmented machine learning framework for improving the bidirectional translation between molecular structures and mass spectra. Specifically, the work investigates how the integration of

graph-based reaction and fragment derivation trees, generated through chemically valid graph transformation rules, can serve as structured and interpretable intermediate representations that enhance both predictive performance and model transparency.

Mass spectrometry-based structure elucidation represents an inherently ill-posed inverse problem: structurally distinct molecules may yield partially overlapping spectral signatures, while minor structural modifications can induce substantial spectral changes. This thesis therefore introduces fragment-level intermediates derived from rule-based graph transformations to inject chemically grounded structure into the learning process.

Instead of directly learning a mapping between molecular graphs and spectral representations, the proposed framework decomposes the task into a structured pipeline in which molecular graphs are transformed into chemically plausible fragment sets, which in turn mediate the relationship to spectral outputs. These fragment sets are embedded into a shared latent space together with molecular and spectral representations, enabling aligned embeddings across the domains.

Ultimately, this thesis aims to demonstrate that incorporating chemically valid graph-based fragment derivations into neural learning pipelines, enhancing both the predictive power and the interpretability of bidirectional structure-spectrum modeling in mass spectrometry and enabling de novo molecular generation from spectral data.

1.3 Scope and Limitations

This thesis is restricted to electron ionization (EI) mass spectrometry, as EI fragmentation reactions are comparatively well understood and mechanistically characterized. [11, ch. 4& 8] This allows the use of chemically informed rule-based graph transformations to generate fragment derivation trees under consistent assumptions. Other ionization techniques are excluded due to their differing fragmentation behavior and the additional modeling complexity they would introduce.

A major limitation arises from the computational cost of graph-based fragment generation. Rule-based graph transformations lead to combinatorial growth in possible fragments, [12, p. 4] especially when multiple reaction rules and recursive fragmentation steps are considered. Ensuring chemical validity and avoiding duplicate graph states further increases computational demands. [13, pp. 2-3]

Due to this scalability constraint, the experimental evaluation is limited to order of magnitude of 100 compounds. While sufficient for demonstrating methodological feasibility and analyzing interpretability, this restricted dataset limits chemical diversity and does not allow

for generalization. Consequently, the results should be interpreted as proof-of-concept validation rather than evidence of large-scale applicability.

Also to be noted is that the reactions are in terms of graph symbols ambiguously defined, and the transformation rules are not exhaustive of all possible fragmentation pathways. This means that the generated fragment derivation trees may not capture all relevant chemical transformations, and the model’s performance may be sensitive to the specific choice of transformation rules. The interpretability analysis is therefore constrained by the assumptions embedded in the rule set, and may not generalize to fragmentation processes that deviate from these assumptions.

Chapter 2

Background and Related Work

2.1 Fundamentals and Capabilities of Mass Spectrometry

Mass spectrometry is a core analytical technique that characterizes chemical species by converting them into gas-phase ions and measuring their mass-to-charge ratio (m/z). [14, p. 5] A mass spectrometer therefore acts as a highly sensitive "molecular balance" whose output – a mass spectrum – reports which ionic species are present and with what relative abundances.

Conceptually, an MS experiment comprises three main steps:

- (i) **ionization**, in which neutral molecules are transferred into the gas phase and charged; [14, p. 1]
- (ii) **mass analysis**, in which ions are separated according to m/z ; and
- (iii) **detection**, in which ion signals are recorded and assembled into a spectrum. [14, p. 5]

Because many compounds form multiple ion species (e.g., adducts [14, p. 21], isotopologues [1, p. 67], fragments), the measured spectrum is not a direct image of the neutral molecule but a structured measurement that must be interpreted in the context of the ionization and fragmentation processes used. [14, p. 16-18]

A key capability of MS is the *identification of unknown compounds*. Even without prior knowledge of the analyte, MS provides constraints that narrow the space of possible identities. Accurate m/z measurements can restrict candidate molecular formulas; isotope patterns (e.g., from ^{13}C , ^{37}Cl , ^{81}Br) further refine these candidates by providing diagnostic abundance ratios and characteristic spacings. [1, p. 263] When fragmentation data are available – either through tandem MS (MS/MS) or through ionization methods that induce fragmentation – structural hypotheses can be tested by matching observed fragment ions and neutral losses to expected

substructures. [1, p. 225] In practice, unknown identification often combines these orthogonal cues with computational searches in spectral libraries and structure databases, where a measured spectrum is matched against reference spectra or predicted fragmentation patterns. [15, p. 8] For complex samples, chromatographic coupling (GC-MS or LC-MS) adds retention time as an additional dimension, [1, p. 475] improving selectivity and reducing confounding effects from co-eluting compounds. [14, p. 233-237]

MS is equally important for the *quantification of known substances*, where the goal is to estimate analyte concentration rather than molecular identity. [14, p. 260] Quantitative MS is grounded in the relationship between the number of detected ions and the amount of analyte introduced into the instrument. [14, p. 261 & 265-266] In idealized settings, ion signal intensity scales proportionally with concentration; in real measurements, this proportionality is skewed. Consequently, reliable quantification typically relies on calibration strategies. *External calibration* constructs a response curve from standards of known concentration, while *internal standardization* – often using isotopically labeled analogs – compensates for variability during sample preparation, injection, and ionization. [14, p. 264-268] The outcome is a method that can measure trace concentrations across wide dynamic ranges, which is central in many applications.

Beyond identity and amount, MS provides a systematic route to *deduce structural features of molecules*. Structural inference is primarily enabled by fragmentation behavior: when a precursor ion breaks apart, the resulting fragment ions reflect chemically plausible bond cleavages and rearrangements, thereby encoding information about functional groups, substructures, and connectivity. [16, p. i49] Certain fragmentation pathways are characteristic – for example, losses of small neutrals such as H_2O , NH_3 , or CO can indicate the presence of hydroxyl, amino, or carbonyl functionalities, while specific fragment series can suggest alkyl chain lengths or aromatic substitution patterns. [1, p. 504] Importantly, structural elucidation from MS data is rarely a single-step deduction; rather, it is an inference problem in which multiple candidate structures may explain the same spectral evidence, especially for isomers. [17, p. 53] As a result, modern workflows increasingly integrate MS measurements with computational models that generate and rank candidate structures based on spectral consistency.

2.2 Principles of Mass Spectrometry

Instrument workflow

Figure 2.1 shows steps in a mass spectrometer: a *sample* is introduced, converted into gas-phase ions in the ion source, separated according to their mass-dependent motion in the *mass analyzer*,

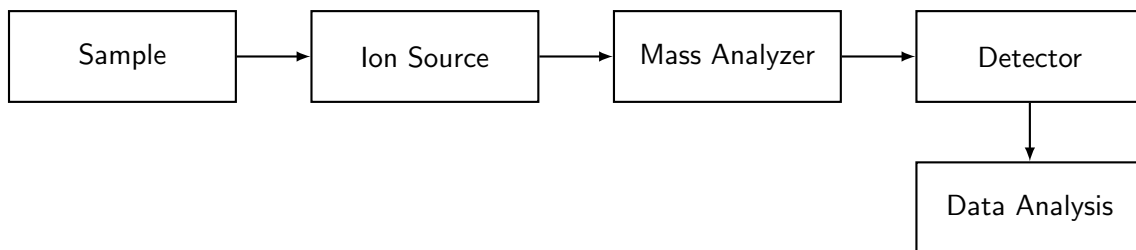


Figure 2.1: Flowchart of a mass spectrometer

[18]

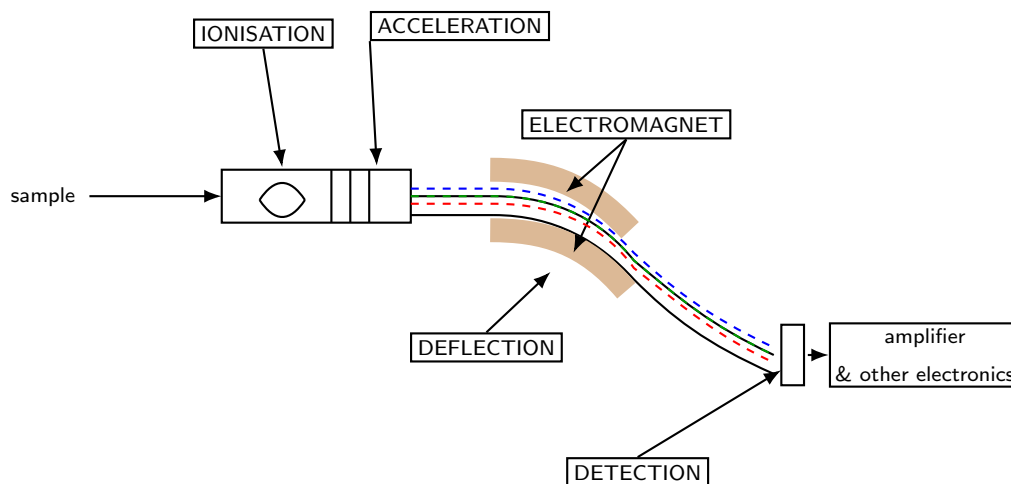


Figure 2.2: Schematic of a mass spectrometer

[19]

detected in the *detector*, and finally interpreted during *data analysis*. The central observable is the mass-to-charge ratio m/z . A mass spectrum is therefore a distribution of signal intensity as a function of m/z , where the signal reflects the abundance of ions reaching the detector. [20, p. 1566] In most practical contexts, intensities are reported as relative intensity, normalized such that the most intense peak equals 100% of the base peak. [20, p. 1550] The x-axis is conventionally labeled with m/z (mass-to-charge ratio), a dimensionless quantity. Labeling with mass units such as *daltons* (Da), amu, or unified atomic mass unit (u) is discouraged – although common in practice – as it causes confusion with multiple charged ions. [20, p. 1516]

Ion notation and precursor concepts

Ions observed in mass spectrometry are typically derived from a neutrally charged molecule M . In electron ionization (EI) a hard ionization method, removal of an electron yields the molecular ion [20, p. 1541] – also called the parent ion – denoted $[M^+]$, [20, p. 1582] a radical cation carrying both a positive charge and an unpaired electron. More generally, the term *precursor ion* refers

to the ion selected for fragmentation, [20, p. 1578] from which product ions are generated. In soft ionization methods like electrospray, ion formation is often dominated by proton transfer, producing a protonated molecule $[M + H]^+$. [20, p. 1580]

Peak types: isotope, fragment, and base peaks

A measured spectrum contains several conceptually distinct peak types. *Isotope peaks* arise from naturally occurring heavier isotopes (e.g., ^{13}C , ^{37}Cl), producing characteristic $M + 1$, $M + 2$, ... patterns adjacent to the monoisotopic ion peak. *Fragment peaks* arise when precursor ions undergo dissociation, yielding smaller product ions whose m/z values and relative intensities encode structural information. The *base peak* is the most intense peak in the spectrum [20, p. 1524] and is used for normalization; [20, p. 1550] it is not necessarily the molecular ion and often corresponds to a particularly stable fragment. [21, p. 2] [1, p. 263]

2.2.1 Ionization and Fragmentation

Ionization techniques are broadly classified into two categories based on the energy imparted to the analyte: *hard ionization* methods, which deliver high internal energy and promote extensive fragmentation, [20, p. 1547] and *soft ionization* methods, which minimize fragmentation and favor molecular ion preservation. [20, p. 1591] [22, p. 34]

Hard Ionization methods

Electron Ionization (EI) is a hard ionization technique in which a stream of high-energy electrons (typically 70 eV) collides with neutral molecules in the gas phase. [1, p. 197] The collision transfers energy to the molecule, causing the ejection of an electron and formation of a radical cation $[M^+]$ with both a positive charge and an unpaired electron. [1, p. 194] Because the internal energy of the molecular ion typically exceeds the activation barriers for bond cleavage, the $[M^+]$ ion spontaneously fragments into smaller product ions and neutral molecules. [23, p. 3] As a result, EI mass spectra are characterized by rich fragmentation patterns that reflect the molecular structure. [24]

The deterministic nature of EI fragmentation pathways makes electron ionization particularly valuable for structural elucidation. [25] Characteristic losses of small neutral species provide diagnostic clues about functional groups and connectivity. EI is the dominant ionization method in gas chromatography-mass spectrometry (GC-MS) and remains widely used in small-molecule analysis. [26, p. 1]

Spark Ionization (SI) , is a hard ionization technique [27, p. 1], that often utilized to detect trace elements at parts-per-billion (ppb) levels. [28, p. 428] It uses high-voltage up to 100 kV pulsed electric sparks between two electrodes to vaporize, atomize, and ionize conductive or semi-conductive materials. [28, p. 428]

Soft Ionization methods

Electrospray Ionization (ESI) is a prominent soft ionization technique particularly well-suited for coupling with liquid chromatography (LC-MS) and for analysis of large biomolecules. [29] In ESI, a liquid sample is converted into a fine aerosol of charged droplets by applying a high electric potential (typically several kilovolts) between a capillary needle and a counter electrode. [30, p. 3] As the droplets travel toward the mass analyzer, they undergo solvent evaporation and coulombic fission, eventually releasing gas-phase ions into the vacuum of the spectrometer. [30, p. 7] [31, p. 53]

ESI ionization is exceptionally gentle, with minimal or no fragmentation in the ion source. [30, p. 7] This characteristic makes ESI ideal for determining accurate molecular weights of intact molecules. [31, p. 211] Notably, ESI can generate multiply charged ions $[M + nH]^{n+}$ from large molecules, reducing their effective m/z and enabling their analysis on instruments with limited mass range. [31, p. 50]

A curtain gas is often employed to shield the ion path, preventing neutral contaminants and solvent droplets from reaching the mass analyzer while allowing ions to pass through. [14, p. 42] The sequence from charged droplet formation to gas-phase ion release is shown in Figure 2.3.

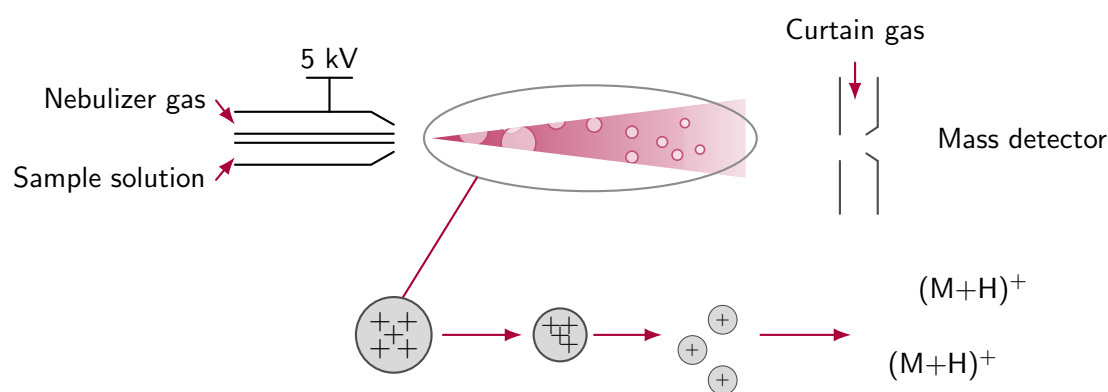


Figure 2.3: Schematic of the electrospray ionization process

[22, p.39]

Atmospheric Pressure Chemical Ionization (APCI) is a soft ionization method used primarily in LC-MS, [22, p. 39] where analytes are ionized at atmospheric pressure via ion-

molecule chemistry initiated by a corona discharge. [20, p. 1522] In contrast to ESI, no high voltage is applied to the inlet capillary; instead, the mobile phase is nebulized, rapidly evaporated in a heated region, [31, p. 56] and the resulting vapor passes a high-voltage needle that generates reagent ions. Solvents dominate the early ionization steps [22, p. 39] and effectively act as a reagent gas, transferring charge to neutral analytes mainly by proton transfer to form $[M+H]^+$.

Because APCI imparts relatively little internal energy, spectra typically show limited fragmentation and are well suited to quantitative LC-MS/MS, [31, p. 171] where signal is concentrated into a few precursor-ion peaks. A heated transfer tube and/or countercurrent curtain gas is used to decluster solvent-analyte adducts prior to ion extraction. [22, p. 39] APCI is often more efficient than ESI for less polar compounds, [1, p. 466] but its higher source temperatures can induce thermal degradation. [22, p. 39]

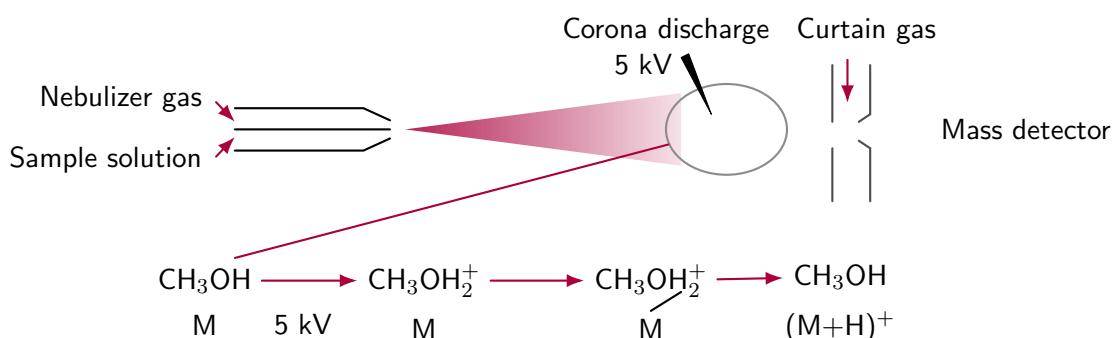


Figure 2.4: Schematic of the atmospheric pressure chemical ionization process
[22, p.39]

MALDI (Matrix-Assisted Laser Desorption/Ionization) [32] is a soft ionization technique [33] that uses a laser to desorb and ionize analytes from a solid matrix. [31, p. 36] It is particularly effective for large biomolecules and polymers, [14, p. 33][31, p. 36] producing predominantly singly charged ions. [14, p. 35]. Tuning laser energy and matrix composition can optimize ionization efficiency and control fragmentation. A general schematic of the MALDI process is shown in Figure 2.5.

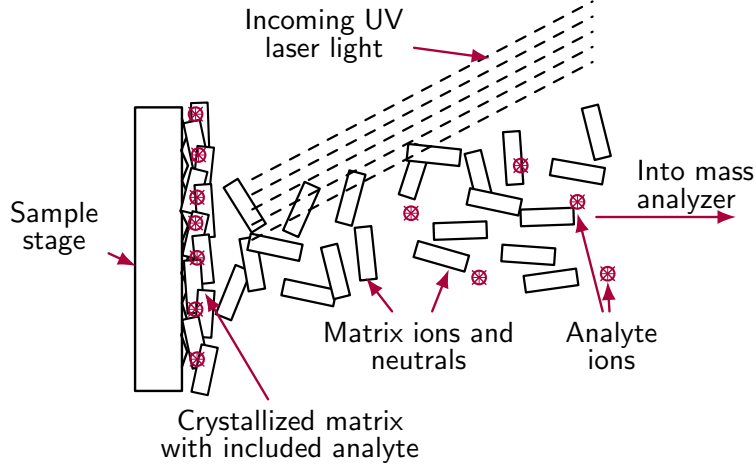


Figure 2.5: UV laser desorbs/ionizes co-crystallized matrix and analyte, creating an ion plume that is sent into the mass analyzer.

[22, p. 42]

2.2.2 Mass Analyzers

After ions are generated in the ion source, they must be separated according to their mass-to-charge ratio m/z before detection. This task is performed by the *mass analyzer*. Mass analyzers exploit the motion of charged particles in electric and magnetic fields to discriminate ions with different m/z values. The motion of an ion with mass m and charge q is described by the Lorentz force equation

$$\left(\frac{m}{q}\right) \mathbf{a} = \mathbf{E} + \mathbf{v} \times \mathbf{B},$$

where $\mathbf{E}$ is the electric field, $\mathbf{B}$ the magnetic field, $\mathbf{v}$ the ion velocity, and $\mathbf{a}$ the acceleration. Because ions with identical mass-to-charge ratios experience identical forces, they follow identical trajectories inside the analyzer. By controlling the fields acting on the ions, species with different m/z values can therefore be separated before reaching the detector. [14, p. 85]

Several types of mass analyzers have been developed. Among the most widely used are magnetic sector analyzers, time-of-flight analyzers, and quadrupole analyzers, each exploiting different physical principles for separating ions.

Magnetic Sector Mass Analyzers

Magnetic sector analyzers separate ions based on their deflection in a magnetic field. When ions enter a magnetic field perpendicular to their velocity, the Lorentz force bends their trajectory into a circular path. The balance between the magnetic force and the centripetal force yields

$$qvB = \frac{mv^2}{r},$$

where r is the radius of curvature of the ion trajectory. Rearranging gives

$$\frac{m}{q} = \frac{B^2 r^2}{2V},$$

assuming the ions were accelerated through a potential V . Thus ions with different m/z values follow trajectories with different radii. By adjusting the magnetic field strength or the acceleration voltage, ions of specific mass-to-charge ratios can be selectively focused onto the detector. Magnetic sector instruments historically played a central role in early high-resolution mass spectrometry. [14, p. 143]

Time-of-Flight (TOF) Analyzers

Time-of-flight analyzers separate ions based on the time required for them to travel a fixed distance after acceleration. When ions are accelerated through a potential V , they acquire kinetic energy

$$\frac{1}{2}mv^2 = qV.$$

Solving for the velocity yields

$$v = \sqrt{\frac{2qV}{m}}.$$

Since ions travel through a field-free drift region of length L , their flight time t is

$$t = \frac{L}{v} = L\sqrt{\frac{m}{2qV}}.$$

Consequently, lighter ions reach the detector earlier than heavier ones. Measuring the flight time therefore allows determination of the mass-to-charge ratio. TOF analyzers are widely used due to their fast acquisition speed and virtually unlimited mass range. [14, p. 126]

Quadrupole Analyzers

Quadrupole analyzers consist of four parallel rods to which a combination of radio-frequency (RF) and direct current (DC) voltages is applied. The resulting oscillating electric field creates ion trajectories that are stable only for a specific range of mass-to-charge ratios. Ions with stable trajectories pass through the quadrupole and reach the detector, whereas ions with unstable trajectories collide with the rods and are removed. By scanning the applied voltages, the

quadrupole sequentially transmits ions of increasing m/z , thereby producing a mass spectrum. Quadrupole analyzers are widely used in routine analytical instruments because of their robustness, compact design, and ability to operate as mass filters or in tandem mass spectrometry configurations. [14, p. 88]

These three analyzer types illustrate the main strategies used in mass spectrometry: spatial separation of ion trajectories (sector instruments), temporal separation based on flight time (TOF), and stability filtering in oscillating electric fields (quadrupoles).

2.2.3 Detection

The dominant detector in EI-MS: Electron Multiplier (EM)

An electron multiplier converts a single incoming ion into a large cascade of electrons (Principle of operation):

1. An ion hits the first dynode
2. Secondary electrons are emitted
3. These electrons are accelerated to the next dynode
4. The process repeats (10-20 stages)
5. Result: a measurable current pulse

2.2.4 Manual Interpretation of the MS Spectrum

Data analysis transforms the raw spectrum into structural hypotheses: isotope patterns constrain elemental composition, while fragment ions and neutral-loss relationships constrain functional groups and connectivity. Importantly, this inference is rarely a single-step deduction; instead, multiple candidate structures can be compatible with the same spectral evidence, particularly for isomers.

Here we are looking at the manual analysis of the mass spectrum
usually well working heuristics

1. What to look for first Molecular ion peak for EI: $[M^+]$
2. look for characteristic isotope patterns

Table 2.1: sample isotope patterns

Element	Diagnostic pattern
Cl	3:1 doublet (M, M+2)
Br	1:1 doublet
S	small M+2 peak
C	^{13}C gives predictable M+1

3. look for base peak (100%) which is the energetically favored fragmentation
4. Analyze fragment differences - look for neutral losses
5. Match fragmentation to chemical logic (reaction rules)

2.2.5 Reactions in Electron Ionization

EI reactions are well known [34]

2.3 Computational Mass Spectrometry

2.3.1 Library Search

is a spectrum-matching procedure used to identify an unknown compound by comparing its measured spectrum against a reference database of known spectra.

one compares: m/z values, relative intensities

workflow: Pre-processing (Noise filtering, Peak picking, Intensity normalization, Optional deisotoping), Candidate filtering (precursor $\pm 5\text{ppm}$), scoring and ranking

2.3.2 In Silico Fragmentation

predicts how a molecule will fragment in a mass spectrometer and generates a theoretical spectrum

workflow: Structure - simulated fragments - predicted spectrum - spectrum matching

2.3.3 De Novo Structure Prediction

in general: from first principles, without relying on experimentally determined templates in MS: inferring the molecular structure directly from MS data, without using a reference database or known templates

2.4 Graph-Based Molecular Representations

2.4.1 Molecular Graphs

Formal definition

A molecular graph is typically defined as

$$G = (V, E) \tag{2.1}$$

where:

- V ... Vertices represent atoms
- E ... Edges represent chemical bonds between atoms

This structure captures which atoms are connected to which, but not their explicit 3D positions unless extended.

Each node may carry attributes such as:

- Element type (C, N, O, ...)
- Formal charge
- Aromaticity
- Hybridization
- Isotope

Each edge may encode:

- Bond order (single, double, triple)
- Aromatic bond
- Bond stereochemistry (E/Z)
- Directionality (important for reactions)

By default, a molecular graph does not include:

- 3D coordinates
- Bond lengths or angles
- Conformation
- Electronic wavefunctions

These require augmented graphs or separate geometric representations.

2.4.2 Chemical Graph Transformation Systems

reaction center (moel, coloured edges and vertecies)

2.4.3 Double Pushout Graph Rewriting

morphisms

$$\begin{array}{ccccc} L & \xleftarrow{l} & K & \xrightarrow{r} & R \\ m \downarrow & & \downarrow & & \downarrow m^* \\ G & \xleftarrow{l^*} & D & \xrightarrow{r^*} & H \end{array}$$

Insertion pushout and deletion pushout

2.4.4 Derivation Graph

or Reaction Networks

describe

2.4.5 Term Mode

Term mode is a representation mode in which chemical structures and reactions are expressed as formal terms (tree-like expressions) rather than as explicit graphs. In this mode, molecules are handled as algebraic objects governed by rewrite rules.

A term is a symbolic expression built from:

- Function symbols (e.g., constructors such as C, O, bond, ring)
- Arguments (subterms)

Conceptually, this is similar to expressions in term-rewriting systems or functional programming.

2.5 Machine Learning in Mass Spectrometry

2.5.1 Deep Learning and Spectral Embedding

CGN GNN, featurizer vs encoder vs embedding, regularization, augmentation

MassGenie: A Transformer-Based Deep Learning Method for Identifying Small Molecules from Their Mass Spectra. <https://doi.org/10.3390/biom11121793>

2.6 Interpretability in Chemical Machine Learning

Chapter 3

Data and Representation

3.1 Datasets

3.1.1 Public Databases

MassBank, pubChem, NIST

3.1.2 Custom Dataset Construction

3.2 Preprocessing - Generating the Synthetic Data

`mod` Derivation Graph reason, goal: gives info of parents of fragments

3.2.1 `mod` Strategy

Forward strategy builds a pipeline that

1. seeds the derivation graph with the molecule via `mod.addSubset`,
2. applies exactly one ionization step wrapped in `sub_group` to constrain it to the current derivation graph, and then
3. applies up to `frag_repeat` fragmentation steps,
4. but only keeps paths whose intermediate fragments satisfy both mass bounds and charge bounds via nested `amu_bound` and `charge_bound` filters.

Backward strategy

1. seeds the derivation graph with the provided fragment graphs via `mod.addSubset`,

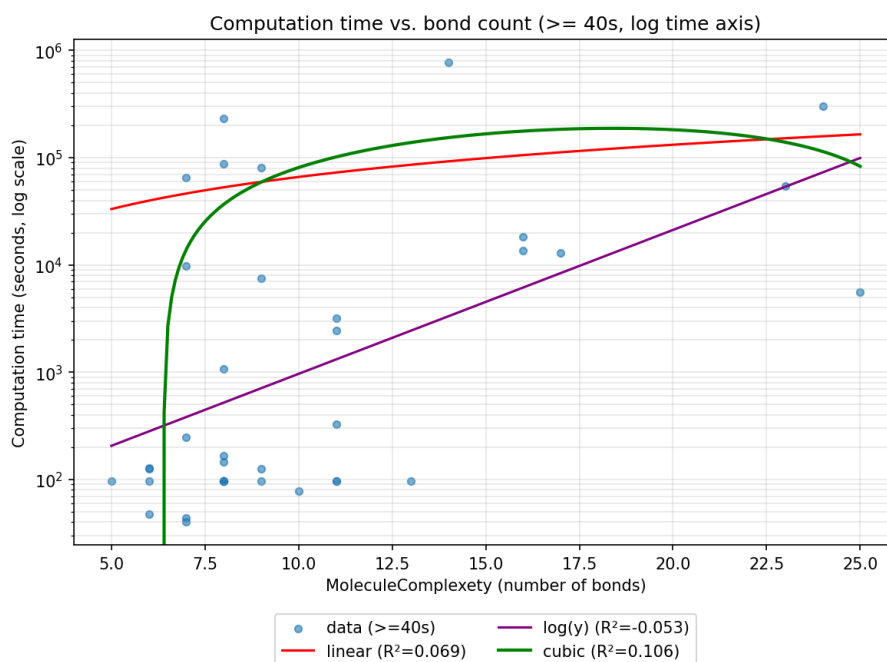


Figure 3.1: Regression of Computation time, filtered of molecules that took at least 40sec to compute

2. then applies up to `frag_repeat` fragmentation rules scoped to the derivation graph,
3. while filtering intermediates to keep only fragments within the given mass window and valid charge bounds (nested `amu_bound` inside `charge_bound`).

3.2.2 computation times

an improve is needed to make this aproach feasible also this is only a part of the rules that are known the compute time per molecule will rise even more

regression analasis, performed

assumption:

- Linearity
- Homoscedasticity
- Independence of Errors

threshold of 40sec as the computation might not be very good

3.3 Latend Representation Schemes

3.3.1 Molecular Graph Encoding

py-geometric

3.3.2 Fragment Graph Representation

technically the same as the Molecular Graph Vertex is a Atom with attributes like element type
Edge is a Bond with attributes bond strengh half bond strength is aromaticity

3.3.3 Spectral Vector and Embedding Representation

latend represantation of the ms spectra

3.3.4 Fragment–Spectrum Mapping

How spectra and fragments are aligned. The “fragment graph” concept - linking structure and spectral space.

Chapter 4

Model Architecture

4.1 Overview of the Learning Framework

4.2 Forward Model: Structure-to-Spectrum Prediction

4.2.1 GNN-Based Encoder

4.2.2 Fragment Predictor or Spectral Generator

4.3 Backward Model: Spectrum-to-Structure Reconstruction

4.3.1 Reinforcement Learning Environment

4.3.2 Reward Design and Shaping

4.4 Bidirectionllity

self supervised learning

4.5 Integration of MØD Hypergraph Derivations

Chapter 5

Experiments and Evaluation

5.1 Experimental Setup

5.1.1 Data Splits and Training Protocols

5.1.2 Evaluation Metrics

cosine jaccard index tanimoto edit-distances

wasserstein metrics

5.2 Forward Model Evaluation

5.2.1 Spectral Similarity and Fragment Prediction

5.3 Backward Model Evaluation

5.3.1 Structure Reconstruction and Validity

5.3.2 Reward Convergence and Exploration Efficiency

5.4 Ablation Studies

without fragment parent awareness, or in other words not using the derivation graph

5.5 Comparison with Baseline Models

CFM-ID,

MSNovelist (<https://doi.org/10.1038/s41592-022-01486-3>),

MassGenie (<https://doi.org/10.3390/biom11121793>) ...

Chapter 6

Discussion

6.1 Benefits of Fragment-Aware Modeling

How fragments improve interpretability.

6.2 Challenges

ambiguity, combinatorial explosion

6.3 potential improvements & future work

speed up graph transformation, especially predicates use one class of molecules

6.4 Summary of Contributions

Manuel Uhler

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Appendix A

My appendix

A.1 MØD Rule Sets and Strategy Definitions

A.2 Hyperparameter Tables and Configurations

A.3 Github Repository

<https://github.com/phireiser/mass-spec>